PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: **Backside Alignment Method for 1 cm x 1 cm die with Heidelberg**

Requestor: **Caitlin Chapin** **Date: 2/4/2019**

Non-SNF staff, please complete the following:

Badger ID: **cchapin3**

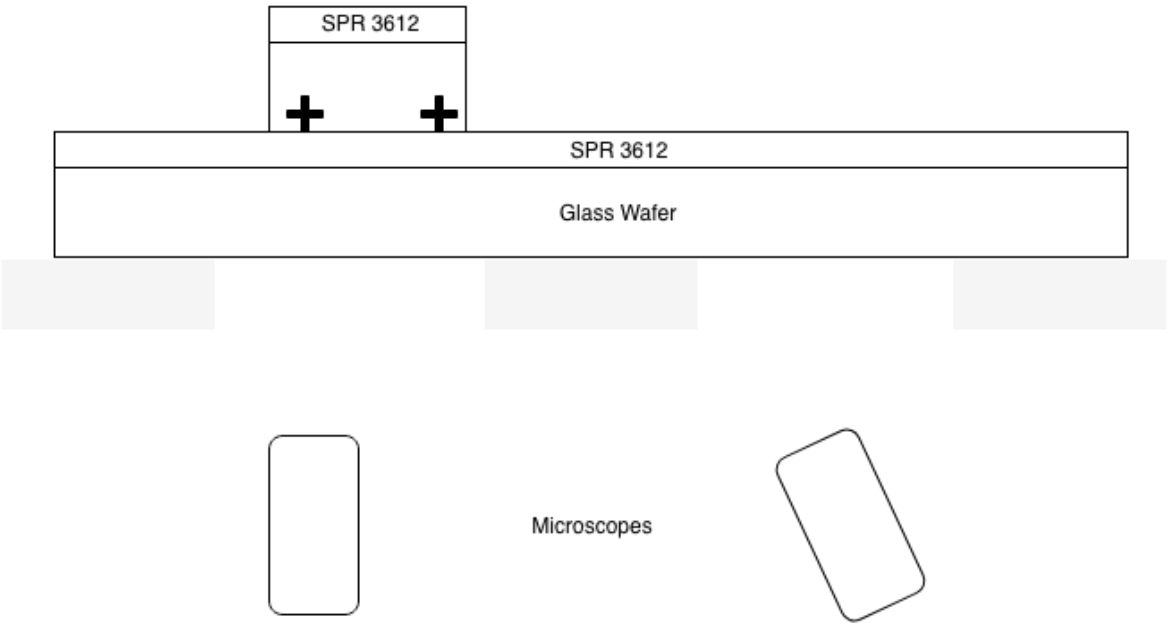
Company name/PI: Debbie Senesky (Extreme Environment Microsystems Lab)

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** No chemical/material new to the lab.
2. **Vendor/manufacturer info:** N/A
3. **Reason for request:**

   We wish to develop a backside lithography processes on 1 cm x 1 cm dies. These are too small reach the edge of the Heidelberg backside microscopes.

   Our proposed solution is therefore to mount the chips on glass wafers (or glass microscope cover slips) with photoresist, and perform a backside alignment with the Heidelberg microscopes.



V1.0

4. **Process Flow:**

- Spin photoresist (SPR 3612) on 1 cm by 1 chip.
- Spin photoresist (SPR 3612) on a 4 ‘’ glass wafer or glass microscope covers.
- Mount the 1 cm x 1 cm piece on top of the wafer, backside facing down.
    - Chip must be placed close enough to center such that one can click the center during sample alignment step, so when heidelberg focuses its on the chip
- Bake the grouped wafer and chip on a hotplate at 90°C for 60-120 s.
- Check for proper adhesion of the chip on the glass (for example by gently shoving the chip with tweezers, if doesn’t wiggle/move with gentle force should be ok).
- Attempt to align the chip crosses using backside lithography on the Heidelberg.
    - For substrate alignment when introducing the sample to the
        - Select “automatic rectangular”
        - Choose the “small” substrate
        - Use the overhead camera to align the objective to the substrate

Alternative process flow:

- Spin photoresist (SPR 3612) on a 4 ‘’ glass wafer.
- Mount the 1 cm x 1 cm piece on top of the wafer, backside facing down.
    - Chip must be placed close enough to center such that one can click the center during sample alignment step, so when heidelberg focuses its on the chip
- Bake the grouped wafer and chip on a hotplate at 110C for 10 minutes.
- Check for proper adhesion of the chip on the glass (for example by gently shoving the chip with tweezers, if doesn’t wiggle/move with gentle force should be ok).
- Spin photoresist (SPR 3612) on mounted chip while mounted to chip.
- Bake the grouped wafer and chip on a hotplate at 90°C for 60.
- Attempt to align the chip crosses using backside lithography on the Heidelberg.
    - For substrate alignment when introducing the sample to the
        - Select “automatic rectangular”
        - Choose the “small” substrate
        - Use the overhead camera to align the objective to the substrate

    - We are still not sure this will work. Will depend on the range of focal length of microscopes of microscopes.

Lab equipment used: SPR 3612, glass wafers/ cover slides, hotplates, tweezers, Heidelberg.

To ensure the heidelberg does not get damaged. Swaroop will join for the first alignment. Additionally, someone will be next to the emergency stop button in case of an emergency.

5. **Amount and form.** N/A
6. **Storage:** N/A
7. **Disposal**: N/A

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: